OLIVER NJERU

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SUMMARY

Software engineer who builds market infrastructure. Built a synthetic frontier stock exchange – matching engine, effective-dated rulebook, settlement, corporate actions, surveillance – deployed as an authenticated service on scale-to-zero infrastructure. Three peer-reviewed publications, one as first author.

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript and TypeScript, SQL, R, Bash.
Backend and infrastructure: FastAPI, Flask, Django; Docker, Google Cloud Run, Firebase and Firestore; REST API design, authentication, rate limiting, metrics and load testing.
Frontend: React, Next.js, Vite; D3.js, ECharts.
Data and ML: pandas, NumPy, SciPy, Parquet; PyTorch, TensorFlow, scikit-learn; multi-agent reinforcement learning (PPO with shared critics), GARCH and stochastic-process simulation.
Domain: limit order books, auctions, market microstructure, transaction-cost analysis, T+N settlement, corporate actions, securities lending, fee and levy engines, AML surveillance.

SELECTED SYSTEMS

Frontier Market Exchange – synthetic frontier equity exchange 2024 – present

- Independent work.
- Price-time-priority limit order book with opening and closing auctions and full level-2 depth. Three matching modes – OTC dealer, periodic call-over, continuous electronic – selected by parameter rather than by branch.
- Effective-dated rule engine: tick sizes, price limits, board lots, sessions, settlement cycles, fee schedules, statutory levies and market-maker obligations resolve against the simulated date. The rulebook is versioned data, which is how production post-trade systems handle the same problem.
- Post-trade: T+N settlement, corporate actions with ex-date tape adjustment, securities lending and short selling, chain-linked index reconstitution that does not jump on listing or delisting.
- Risk engine structured as monitors, policy and mitigations; AML and sanctions surveillance with onboarding, risk scoring and a counterparty relationship graph.
- Seeded, bit-reproducible world generation validated against literature stylized facts as a pass/fail gate. Deterministic replay, hash-chained event log, snapshot and restore, all enforced by test.
- Precompute-first architecture keeps the request path cheap: measured p95 under 50 ms server-side per session step, at effectively zero idle cost on scale-to-zero infrastructure.

Exchange demonstration application – interactive system 2026

- Flask service over the engine with a vendored, hash-pinned charting library; no CDN, no telemetry, no external calls. Exports to a static site and to a single self-contained HTML file that runs with no server or install.
- Includes a simulation lab with counterfactual and multi-seed runs, a world builder, a scenario branch-and-replay tool, a fee-policy sandbox, an operator-approved risk desk, a brokerage and underwriting tier, a multiplayer trading floor with shared sessions, and a historical role-play mode scored against counterfactual replay.
- Building a market-maker quote-presence gauge surfaced three latent engine defects – an uncleared registry, a float comparison deciding spread compliance, and a regulatory clock that never advanced – all diagnosed and fixed.

Platform service – API and web client for the engine 2026

- FastAPI service on Cloud Run: session lifecycle over the engine's snapshot and restore, order entry, portfolio and market-data reads, backtests with risk-adjusted reporting.
- Firebase authentication, per-user rate limiting, latency histograms and per-tenant counters, containerized build and deploy, durable sessions with compressed write-through persistence.
- React and TypeScript single-page client on Firebase Hosting. Full login-to-fill flow verified on the hosted deployment.

EXPERIENCE

Black Eternum Capital LTD, Nairobi February 2024 – present
Founder and Director (August 2024 – present); *Founding Engineer, Replicant Trader* – the company's product arm (February 2024 – present)

- Design and build the Frontier Market Exchange, described under Selected Systems above. Work began February 2024 as a bilateral dealer-to-client request-for-quote market and grew into the exchange system the arm now operates, deployed at replicanttrader.com.
- Kenyan private limited company. Own product, engineering, compliance and governance.
- Built the company's documentation and compliance framework, including a compliance code mapped to Kenyan statute and a pre-launch product compliance gate.

Independent Researcher – Nairobi June 2026 – present

- Multi-agent reinforcement learning and market design.

USIU-Africa, School of Science and Technology October 2023 – May 2026
Machine Learning Research Assistant (May 2024 – May 2026); *Research Assistant* (October 2023 – May 2024)

- Multi-agent reinforcement learning and market design for frontier capital markets: how exchange rules and trading frictions shape participant behavior, market quality and inclusion.
- Implemented a multi-agent PPO framework with a shared critic across thirteen heterogeneous agent classes; modeled fee schedules, price limits and settlement delay to study adaptation and liquidity withdrawal.
- First author of the VINE JIKMS article (2026) from this work. Earlier stint produced the IST-Africa 2024 publication.

Software Engineer Intern – IBM Research Africa, Nairobi June 2023 – August 2023

- Project SPARK: interface letting domain scientists direct biomedical foundation models for drug-candidate generation and screening.
- Built high-fidelity prototypes in Next.js with the IBM Carbon Design System and D3.js. Scrum master for the frontend team. Co-author on the resulting IJCAI 2024 paper.

Junior Software Developer – Mombasa March 2018 – September 2018

- Built secure database software for small local businesses in Microsoft Access and Visual Basic.

PUBLICATIONS

- "SPARK: Harnessing Human-Centered Workflows with Biomedical Foundation Models for Drug Discovery." IJCAI 2024, Demonstrations Track, pp. 8713–8716. Co-author.
- "DbAPI: A Backend-as-a-Service Platform for Rapid Deployment of Cloud Services." IST-Africa 2024. Co-author.
- "Market data governance as knowledge management: managing temporality via rule versioning, provenance and auditability." VINE JIKMS 2026, ahead of print. doi:10.1108/VJIKMS-03-2026-0132. First author.

EDUCATION

BSc Applied Computer Technology, USIU-Africa, Nairobi April 2025
Software Engineering concentration. Graduated with honors. Dean's List, multiple semesters.

Level 4 Diploma, Investment Bank Sales and Trading, AmplifyMe, London March 2023
In partnership with Morgan Stanley; accredited by The London Institute of Banking and Finance. Ranked first in an investment bank sales and trading simulation; role performance 91 percent sell-side, 87 percent buy-side.

Diploma, Business Information Technology – Strathmore University, Nairobi September 2021

SELECTED AWARDS

Vice Chancellor's Grant, USIU-Africa (2022–2025) · High Potential Candidate, AmplifyMe Finance Accelerator (2023) · Team Lead, JP Morgan Chase Code for Good Africa (2023) · Top 25 percent, CME Group FX Trading Challenge (2023) · Junior Achievement Africa Scholarship (2022)